Question:

An 18-month-old boy presents to the emergency department for malaise. The boy’s parents report worsening fatigue for 3 days with associated irritability and anorexia. The patient’s newborn screening revealed a point mutation in the beta-globin gene but the patient has otherwise been healthy since birth. On physical exam, his temperature is 102.4°F (39.1°C), blood pressure is 78/42 mmHg, pulse is 124/min, and respirations are 32/min. The child is tired-appearing and difficult to soothe. Laboratory testing is performed and reveals the following:Serum:Na+: 137 mEq/LCl-: 100 mEq/LK+: 4.4 mEq/LHCO3-: 24 mEq/LUrea nitrogen: 16 mg/dLCreatinine: 0.9 mg/dLGlucose: 96 mg/dLLeukocyte count: 19,300/mm^3 with normal differentialHemoglobin: 7.8 g/dLHematocrit: 21%Mean corpuscular volume: 82 um^3Platelet count: 324,000/mm^3Reticulocyte index: 3.6%Which of the following is the most likely causative organism for this patient's presentation?Answer Choices:A. Haemophilus influenzaeB. Neisseria meningitidisC. SalmonellaD. Streptococcus pneumoniae

Reasoning + Answer:

<think>The patient had malaise, fatigue, and anorexia. The newborn screening revealed a point mutation in the beta-globin gene. The laboratory results showed abnormal hemoglobin and hematocrit levels. Haemophilus influenzae, Neisseria meningitidis, Salmonella, and Streptococcus pneumoniae were possible causative organisms.</think><answer>Haemophilus influenzae</answer>